

Arrive on time, arrive ready

A data-driven travel plan for people with time sensitive needs

Team 7

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Executive summary

The problem

1 in 10

flights gets the traveler there an hour late or not at all (9.7%); 23.5% are disrupted. Most flights land early; the cost sits in a tail that is predictable.

What we found

3x

Evening departures are about three times as likely to be disrupted as early-morning ones. The right airline on a route is worth 5 to 7 points, bad weather nearly doubles the risk, and a stop lifts hour-late arrivals from 9% to 16%.

What we recommend

4 defaults

in the booking tool, not bans: a morning preference for time-critical trips; nonstop first, with 90-minute layovers via approved hubs; the preferred airline by route; a weather buffer in risky months.

The impact

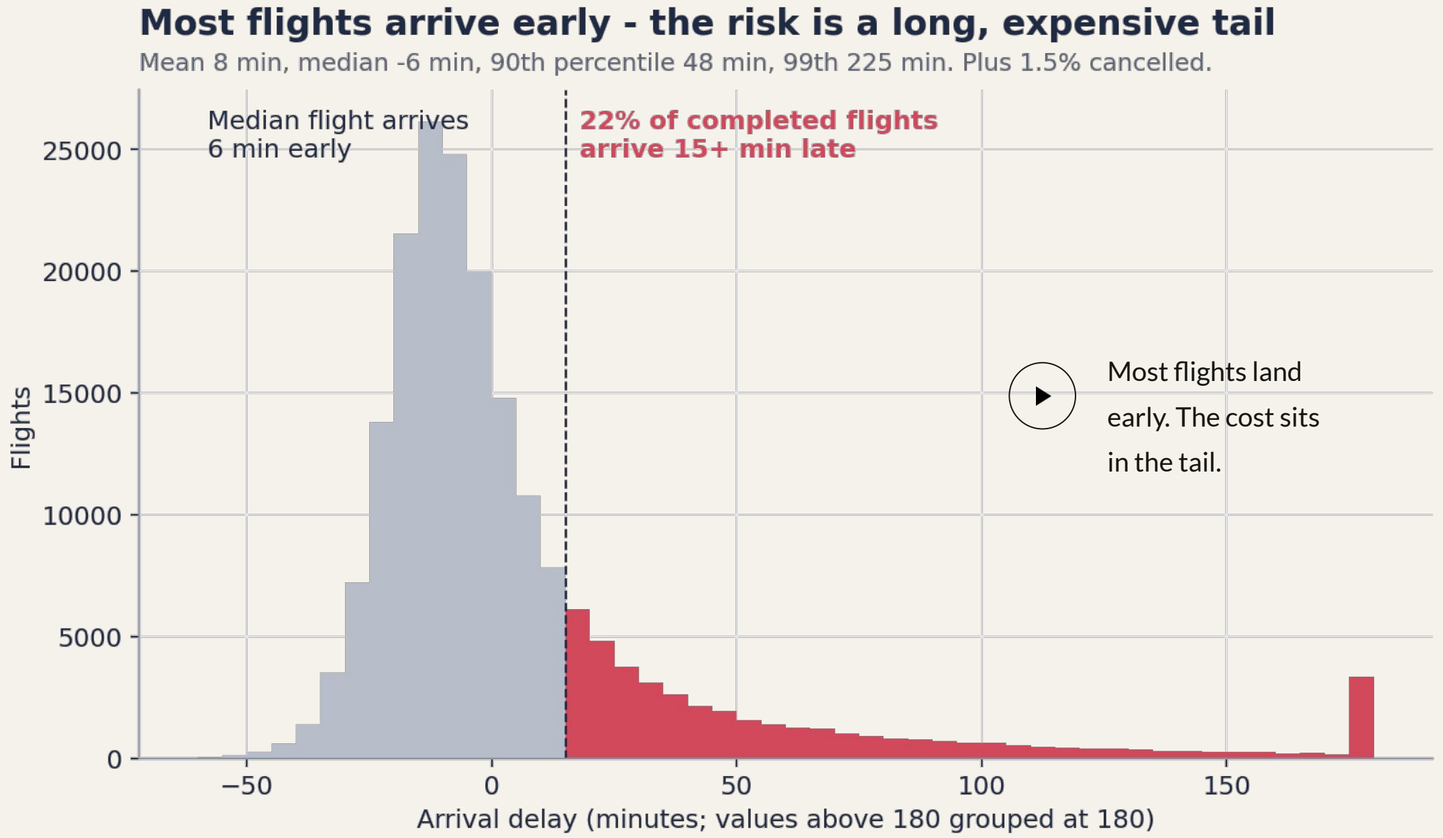
96 → ~50

hour-late trips per 1,000, worth $13k to $76k per 1,000 trips depending on what an hour and a meeting are worth. The ask: a one-quarter pilot in two business units against a control group.

One flight in ten gets you there an hour late

What we did

We took every US domestic flight of ten airlines for a full year, 6.4 million flights, added the weather at both airports, and asked one question: which choices get a traveler to the meeting on time?



23.5%

disrupted: arrived 15+ min late, cancelled or diverted

9.7%

severely disrupted: 60+ min late, cancelled or diverted

1.5%

cancelled (95,166 flights) and 0.3% diverted (18,015)

460 h

of delay per 1,000 trips once cancelled travelers are rebooked (simulation)

Fly in the morning

Morning departures had lower overall disruption rates, with severe disruptions also increasing later in the day.

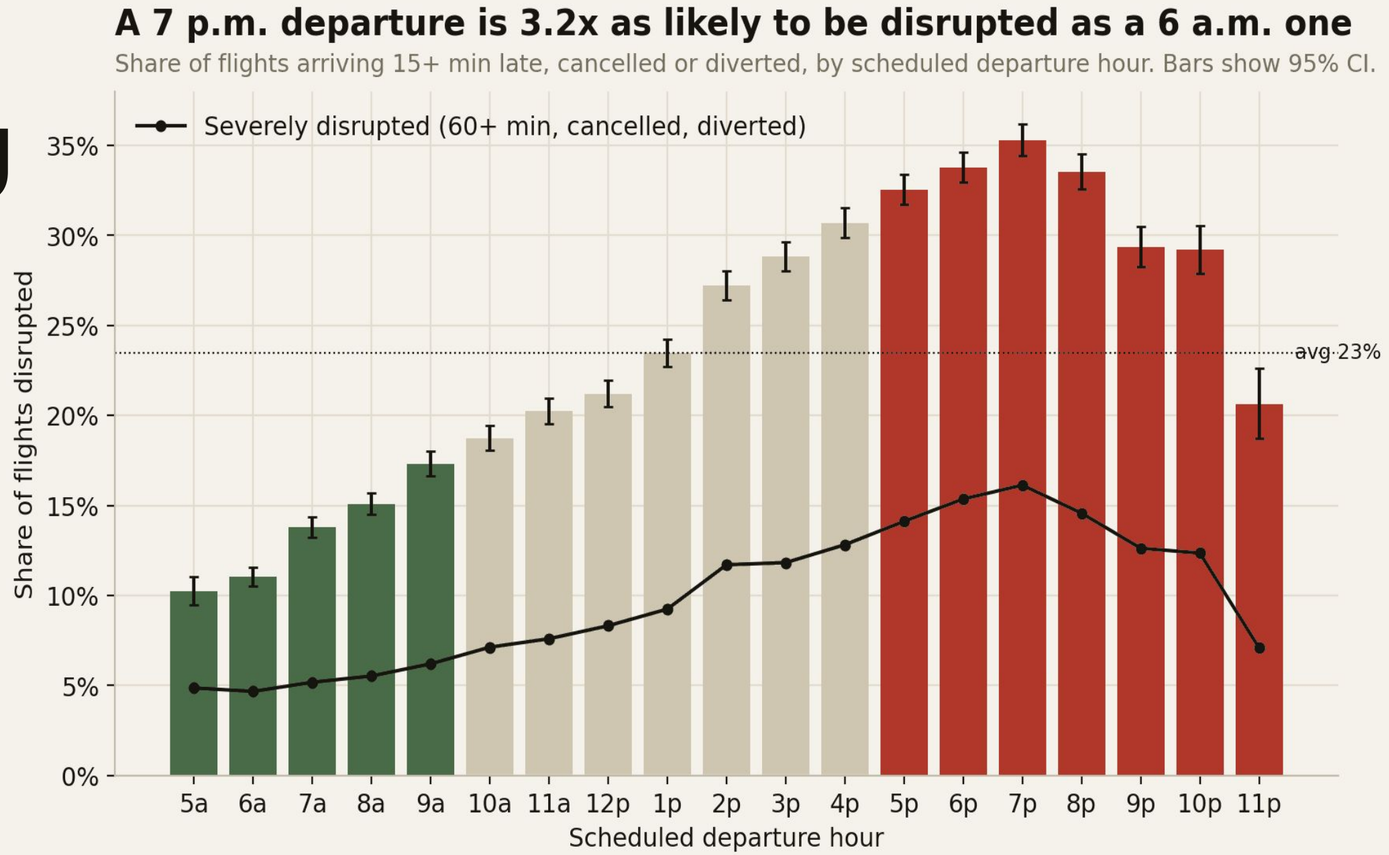
What we found

A 7 p.m. flight had 3.2 times the overall disruption rate of a 6 a.m. flight. Disruption risk generally increased throughout the day, making evening departures less reliable.

What it means

For important business trips, prioritize departures between 7 and 10 a.m. when practical. This offers lower disruption risk without requiring extremely early flights.

Chance of disruption by scheduled departure hour, all flights.



Same airline, same route: evening is worse on 98% of routes.

Pick the airline by route

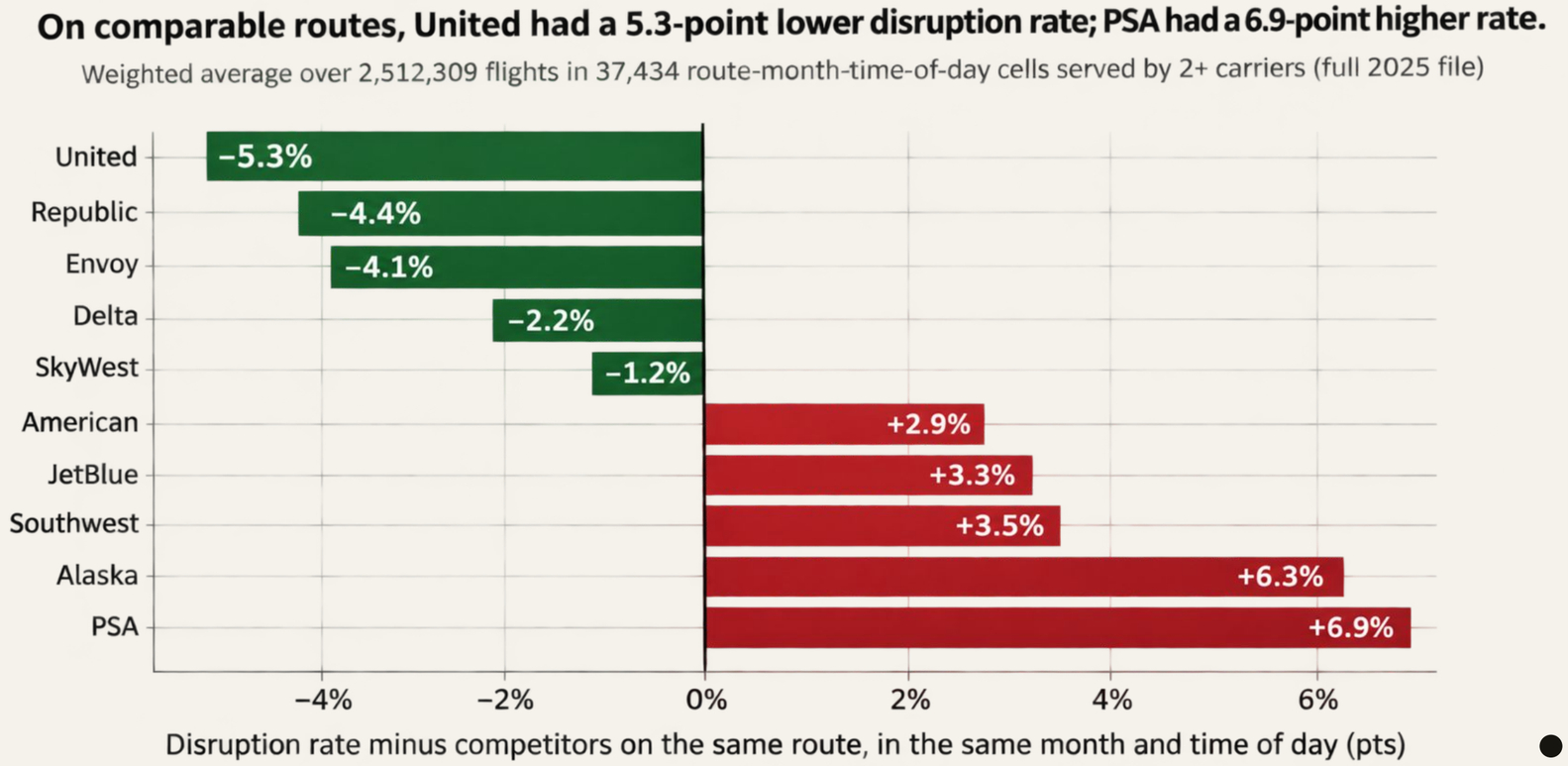
Raw rankings mislead: an airline that flies mostly in calm western skies looks better than it is. We compared airlines only on the same route, month and time of day.

What we found

On comparable routes, months, and departure times, United had a 5.3-percentage-point lower disruption rate than competing airlines. Alaska and PSA had rates 6.3 and 6.9 points higher, respectively.

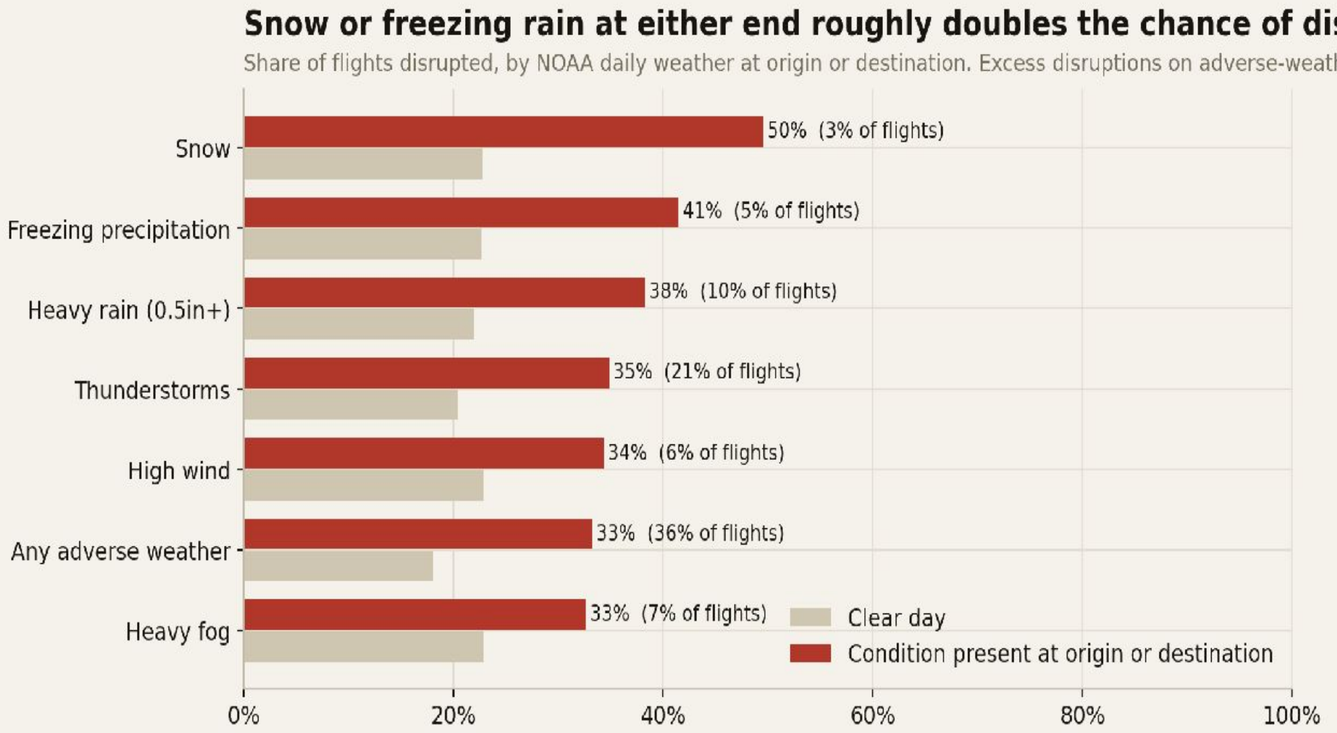
What it means

On routes with multiple airline options (approximately 43% of trips), compare carriers using historical reliability—not brand reputation alone. Recommend lower-risk options when fares and employee schedules are comparable.



Flights touching an airport with bad weather are 1.8x as likely to be disrupted.

Weather increases disruption risk



What we found

Flights experiencing adverse weather had a 33% disruption rate, compared with 18% in clear conditions. Snow and freezing precipitation showed particularly high disruption rates of approximately 50% and 42%, respectively.

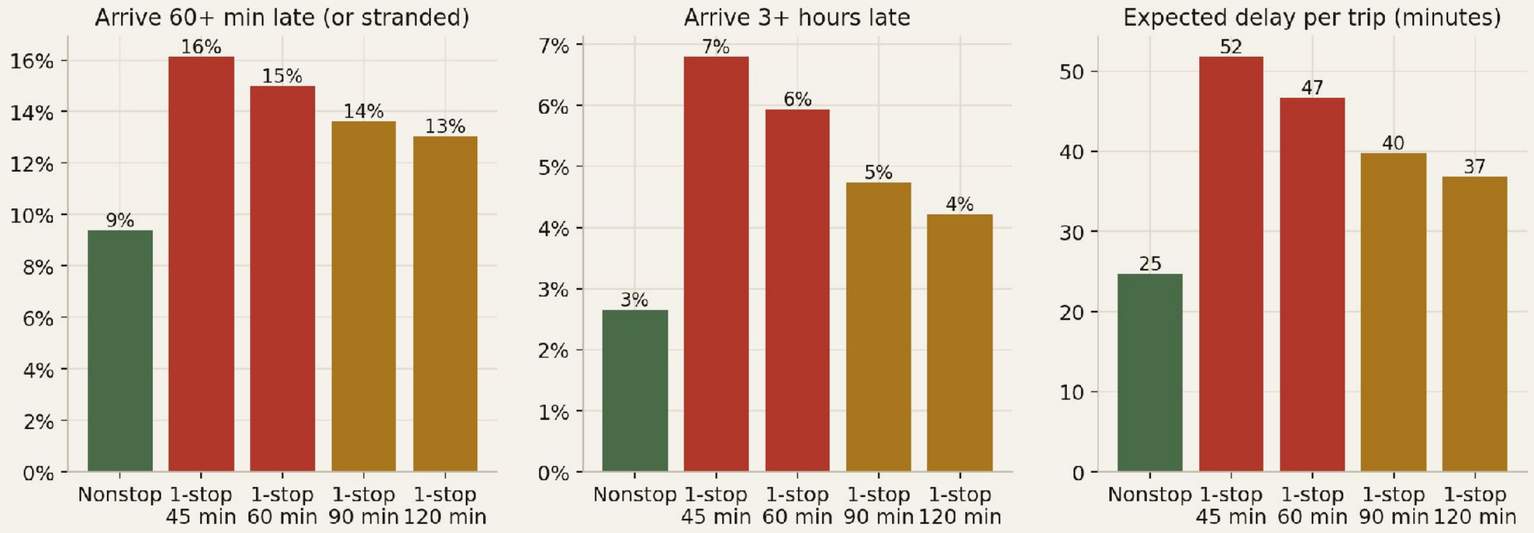
What it means

For time-sensitive business trips, consider seasonal weather risks. Check forecasts 1–3 days before departure and consider earlier flights or day-before travel when worthwhile.

Connections increase travel risk

A one-stop trip is 1.7x as likely to make you an hour late, and longer layovers only partly fix it

1730 city pairs between the 60 busiest airports; one-stops built from real flight pairs via the 20 busiest hubs; cancellations rebooked on the next flight



What we found

Nonstop trips arrive an hour late 9.4% of the time. The same city pairs with one stop: 16%. Overnight strandings triple.

Layover length matters

Raising the minimum layover from 45 to 90 minutes cuts missed connections from 51 to 23 per 1,000 connecting trips.

The hub decides the odds

Connecting through Salt Lake City, Los Angeles or Seattle: about 10% an hour late. Through Washington National, Dallas or LaGuardia: over 20%.

What it means

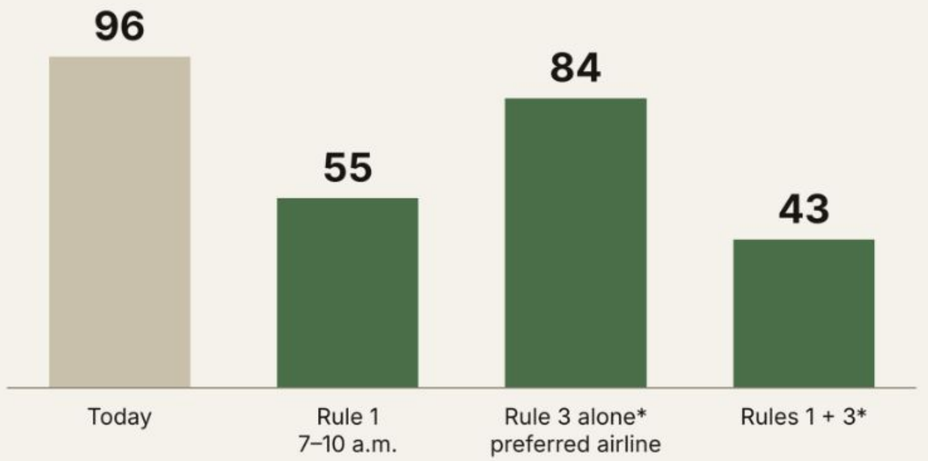
Prioritize nonstop flights for important business trips. When connections are necessary, allow at least 90 minutes and compare the added travel risk against airfare savings.

What we recommend

Four recommendations, shown to travelers as suggestions and alerts, not bans. Modeled on the same routes and volumes as flown, per 1,000 trips, if followed in full. Rule 1 alone is worth about $44k per 1,000 trips at mid assumptions ($13k–$76k across our range).

Trips per 1,000 arriving an hour late or stranded

If every time-critical trip followed the rule. 2025 routes and volumes.



* On the 43% of trips where 2+ airlines fly the route; those trips are 98 per 1,000 as flown.

Rule	What changes	Impact per 1,000 trips
1 · Morning preference	Time-critical trips default to 7 to 10 a.m. departures	Hour-late trips 96 → 55 per 1,000 (−43%)
2 · Nonstop first	Connect only with 90+ minutes via approved hubs	Missed connections 51 → 23 per 1,000 connecting trips
3 · Airline by route	Preferred airline on routes with a real choice	1–15% lower modeled severe-disruption risk on eligible routes; additional benefit when combined with morning preference.
4 · Weather buffer	Day-before travel in risky months; rebook on a bad forecast	Summer/winter afternoon trips 16% → 7% if moved to morning (lowest confidence)

Implementation

Light touch: recommendations, not bans.

Trip Risk Checker

Responsible Party(RP): Analytics Consulting Team + Company Representative. Use Trip Risk Checker to compare flight-disruption risk before employee's book business travel. Recommend reliable departure times, nonstop options, and suitable airlines while considering cost and schedule flexibility. Employees make the final booking decision.

Pre-Trip Risk Alert

RP: Consulting Team + Company Representative. For important business trips, review weather forecasts and travel risks 72 hours before departure. Flag high-risk itineraries and suggest earlier flights, longer connections, or day-before travel when justified by the meeting's importance and additional cost.

Pilot

RP: Consulting Team + Company Representative. Run a 90-day pilot across two business units, with a comparable control group. Track severe disruptions, total travel costs, and employee feedback. Expand only if reliability gains justify additional costs.

Keep it current

RP: Consulting Team. Preferred-airline and approved-hub lists are refreshed every quarter from the public flight data. The refresh pipeline is built.

How we measure success: Compare severe disruptions (60+ minute delays, cancellations, or diversions) per 1,000 trips between pilot and control groups. Also track total travel costs, delay hours, overnight strandings, employee adoption, and traveler satisfaction. Expand only if reliability improves enough to justify additional costs.

Days 0–30 | Prepare

Select pilot business units and routes, set baseline KPIs, and set travel-cost guardrails.

Days 31–60 | Pilot

Introduce flexible booking recommendations, monitor employee adoption, and track reliability and any extra cost.

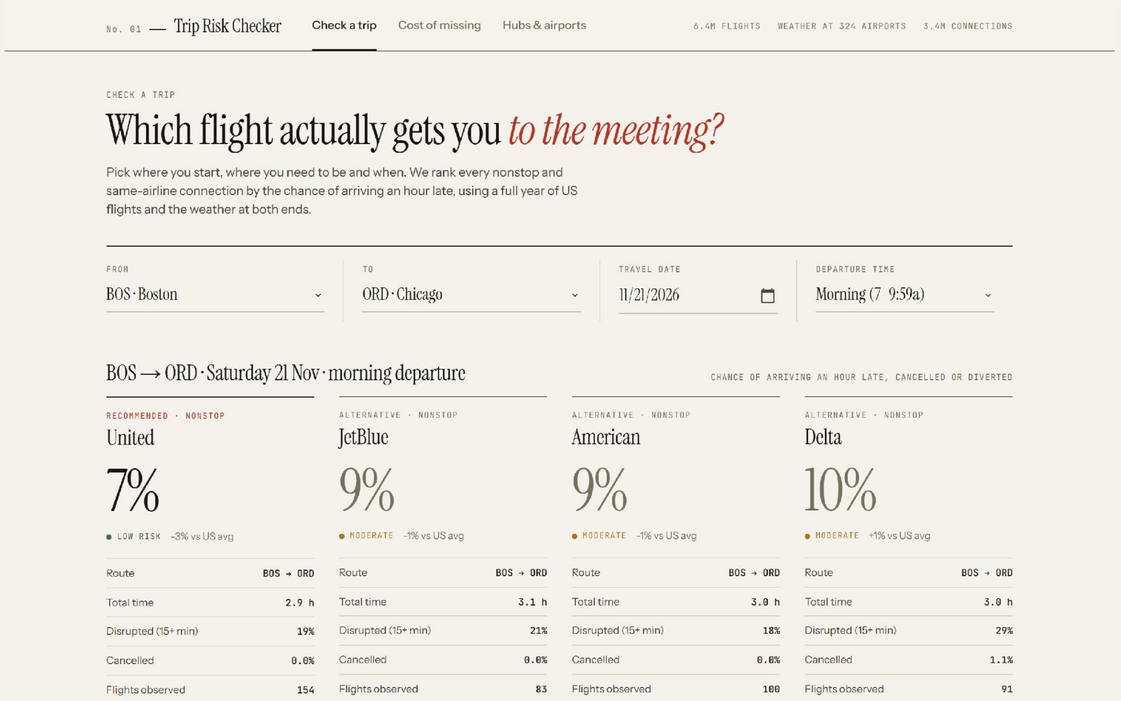
Days 61–90 | Evaluate

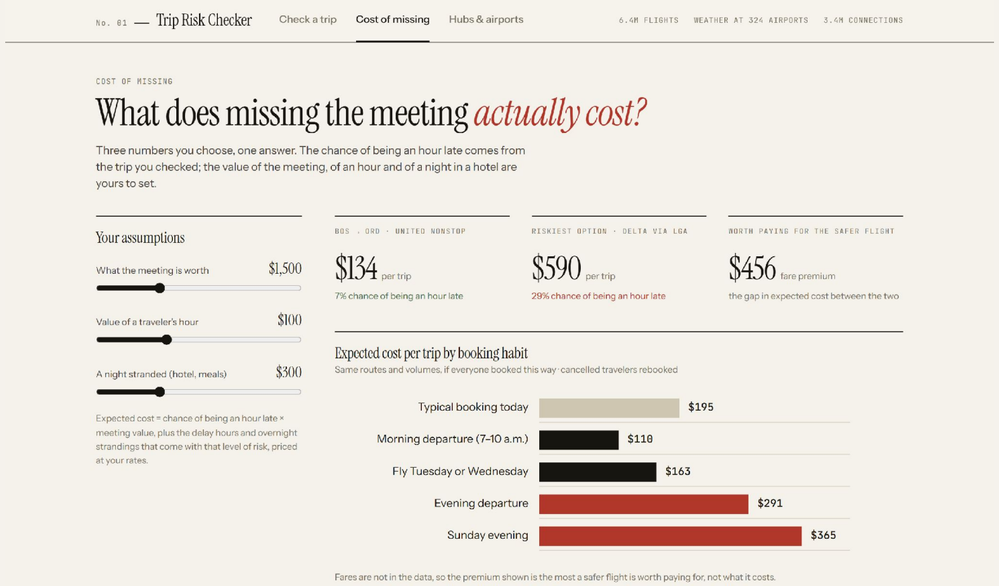
Compare results with a similar control group, assess net business value, and recommend scaling, adjusting, or stopping the policy.

Next step · The website

Check a trip. This tab allows the user to see the risks of taking specific flight routes and which carrier to take. It takes weather, historical data and timings into account to give the final scores.

Cost of missing. Put your own value on the meeting, an hour and a night stranded; see what each option is expected to cost and how much a safer flight is worth paying for.



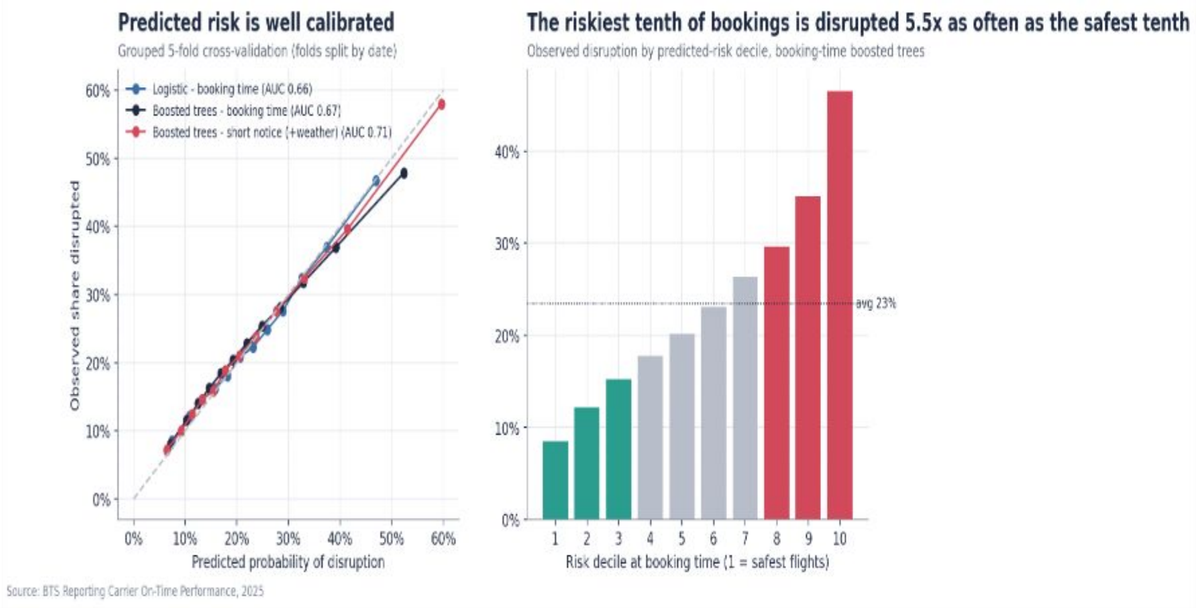


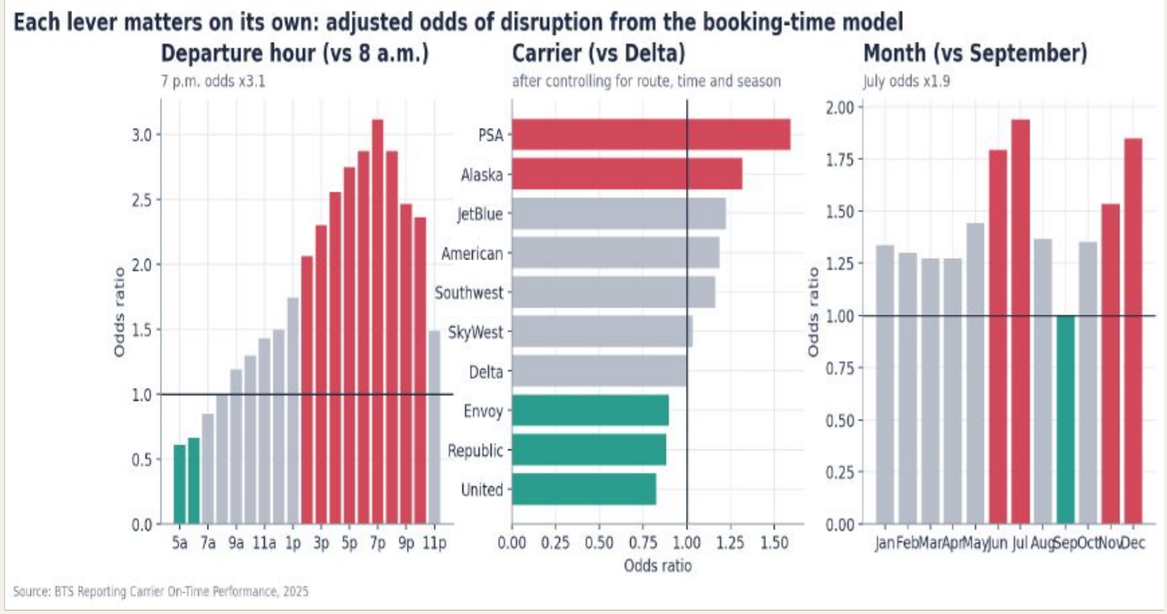
Live at trip-risk-checker.vercel.app.
Hubs & airports page maps disruption, weather exposure and winter fragility.



Thank You

Appendix A1 — Model validation: calibrated at booking time; day-of weather adds real information





Model (5-fold CV grouped by date; 200k sample)	AUC	Avg precision	Brier
Logistic — booking time	0.665	0.375	0.168
Boosted trees — booking time	0.671	0.381	0.168
Logistic — short notice (+ day-of weather)	0.699	0.437	0.161
Boosted trees — short notice (+ day-of weather)	0.706	0.450	0.160
Always predict the base rate	0.500	0.235	0.180

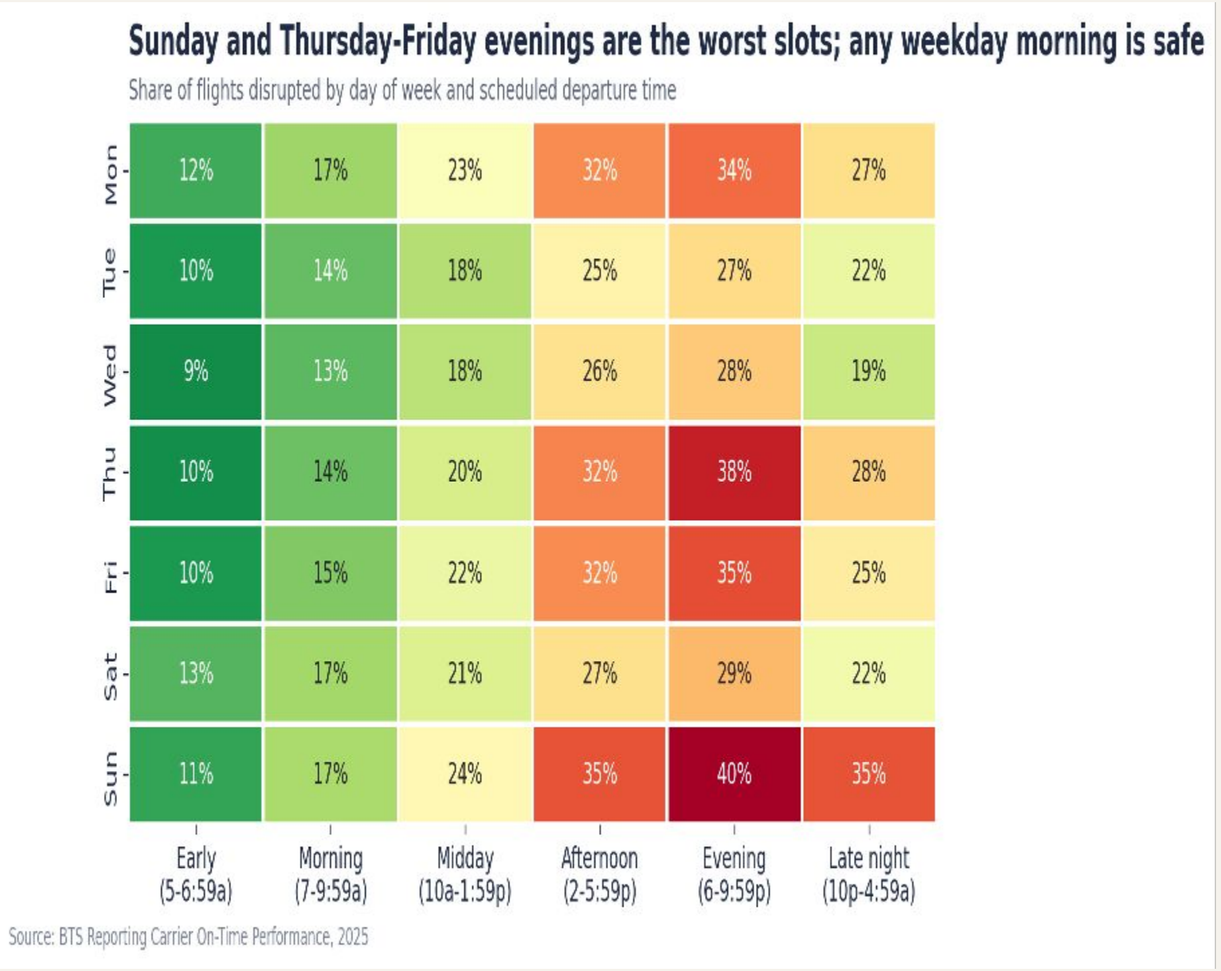
Appendix A2 — Data quality: no rows dropped, every missing value structural, sample reproduces the population

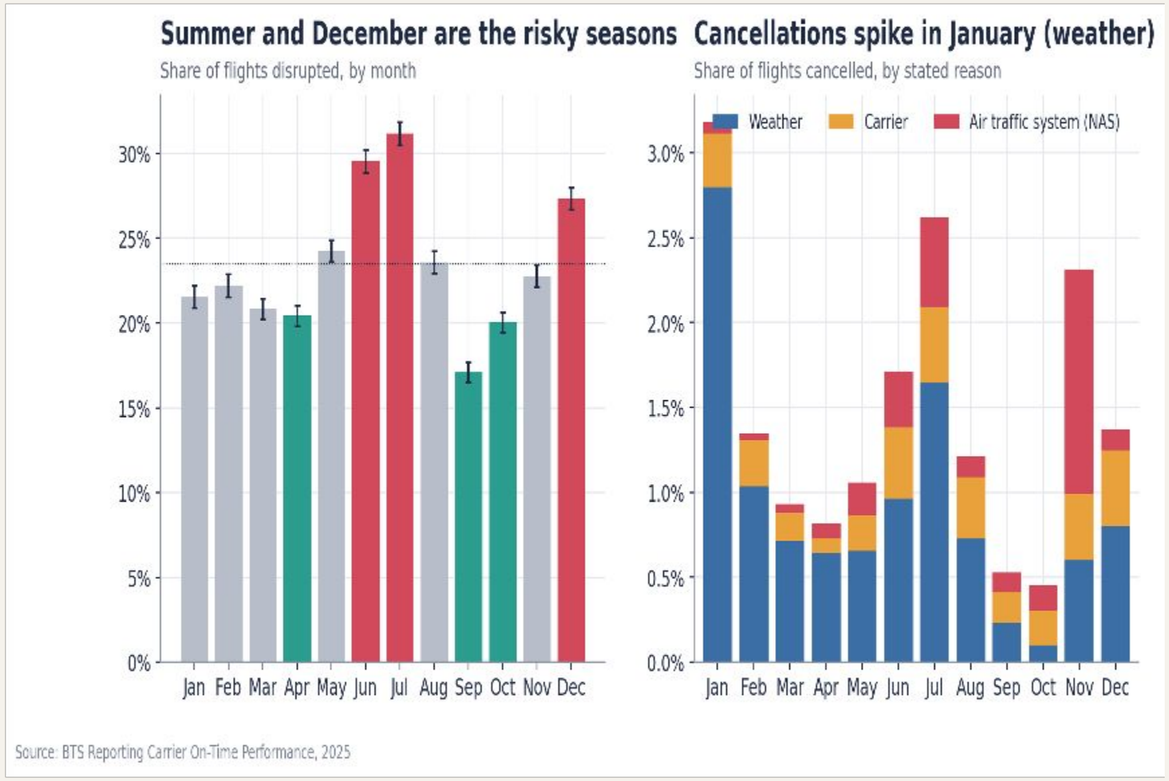
Issue	Rule	Rows (200k sample)	Rows (full file)
Times stored as HHMM numbers	Read as text, convert to minutes after midnight; '2400' -> '0000'	200,000	6,398,056
Arrival date not given	Rebuilt scheduled/actual arrival timestamps from schedule (handles overnight + time zones)	200,000	6,398,056
ARR_DELAY blank for cancelled flights	Kept; counted as disrupted and severely disrupted	2,907	95,166
ARR_DELAY blank for diverted flights	Kept; counted as disrupted and severely disrupted	570	18,015
Delay-cause columns blank when delay < 15 min	Left blank; cause analysis uses delayed flights only	156,488	—
OP_CARRIER duplicates OP_UNIQUE_CARRIER	Dropped duplicate; renamed to CARRIER	200,000	6,398,056
Extreme delays (> 1,000 min)	Kept (real events); report medians/percentiles beside means	91	—
TAIL_NUM blank	Kept; only occurs on cancelled flights; not used as a feature	248	—
Rows removed	None	0	0

Sample vs population

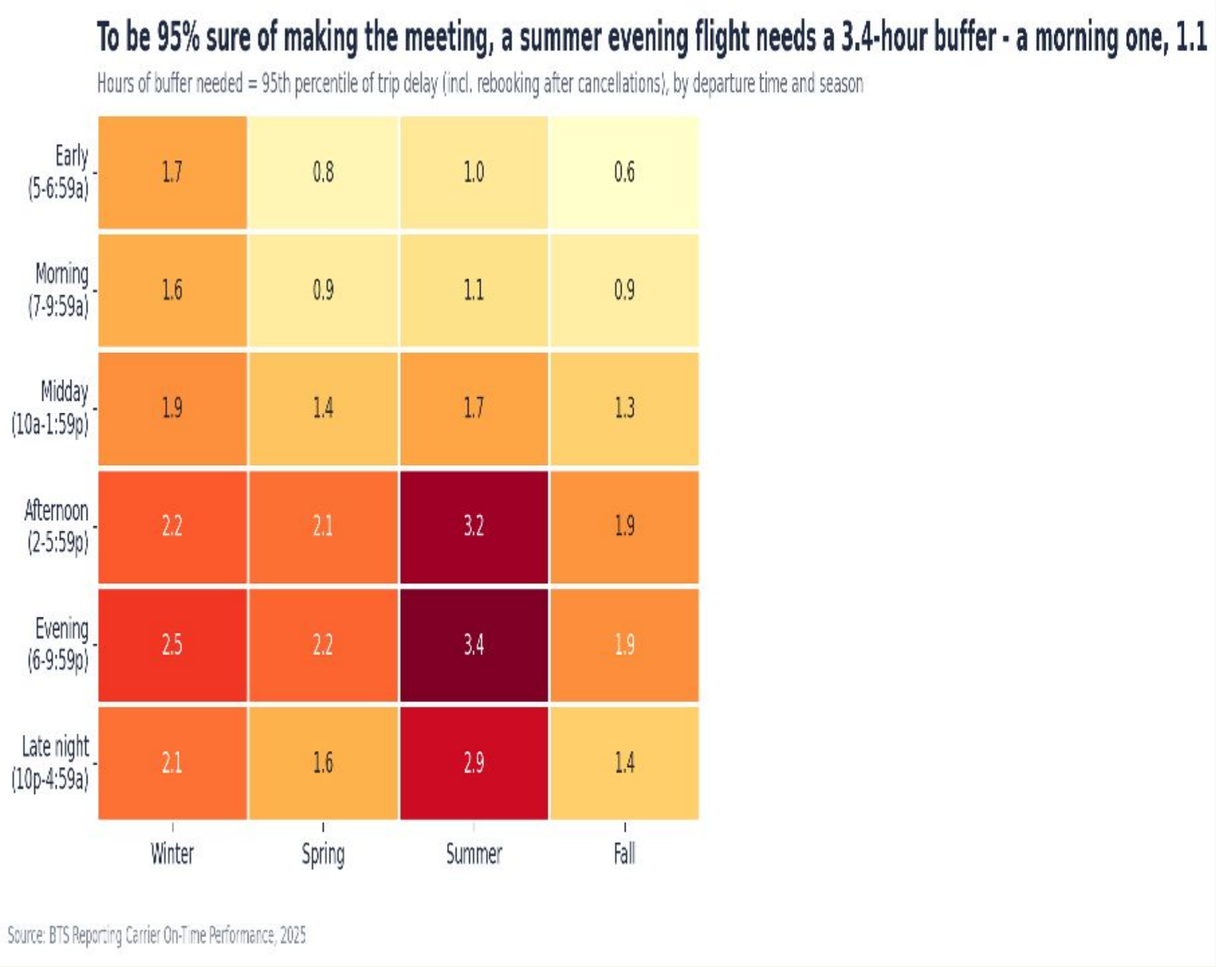
- 200,000-flight uniform sample of 6,398,056 flights (organisers' validation report reproduced exactly: 2,907 cancelled, 570 diverted, 327 airports).
- Full file: 329 airports, 95,166 cancellations (1.5%), 18,015 diversions (0.3%); disrupted 23.55%, severe 9.70%.
- Every carrier's population rate falls inside the sample's 95% Wilson interval.
- Route, hub and connection results use the full file; the sample carries the national patterns and the model.

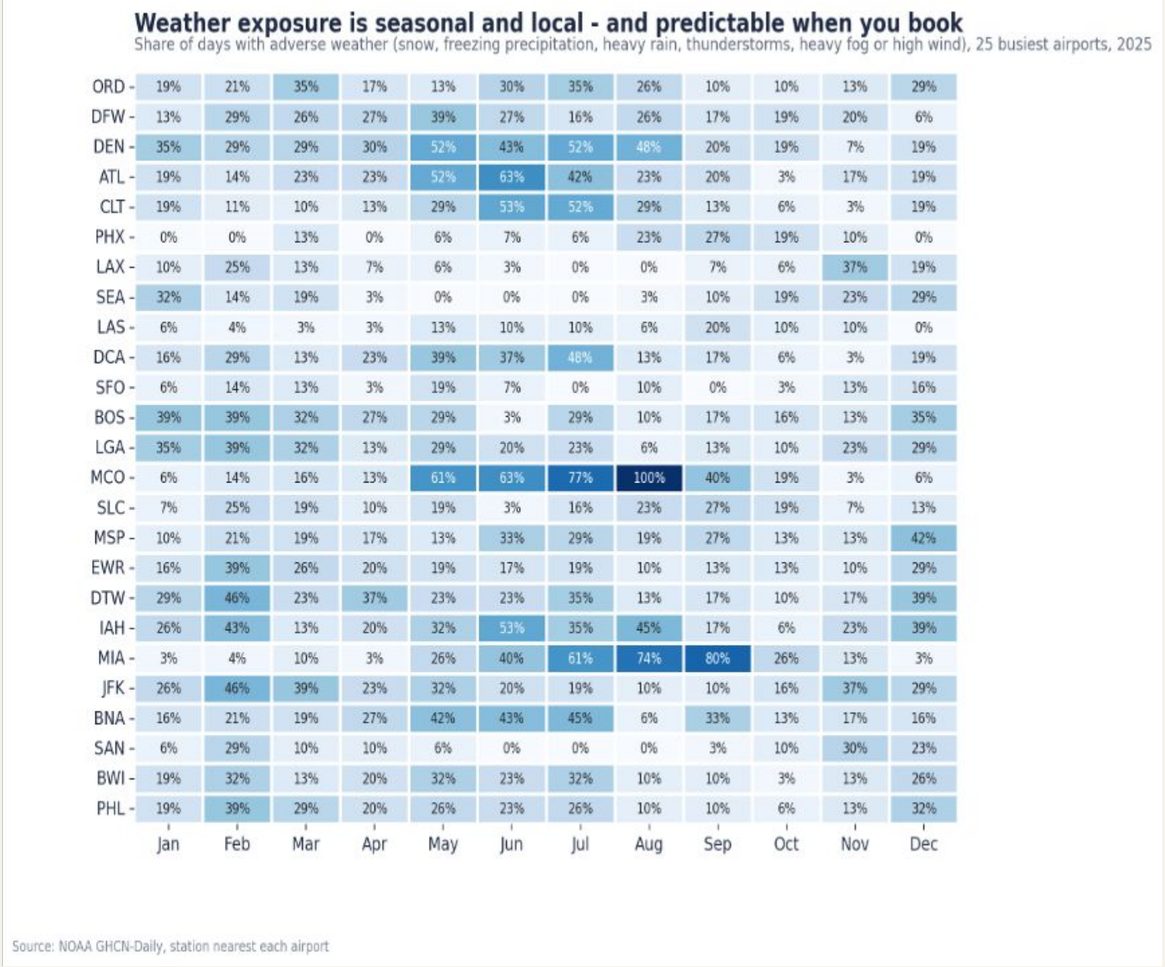
Appendix A3 — When: day-of-week by time-of-day, and the month profile



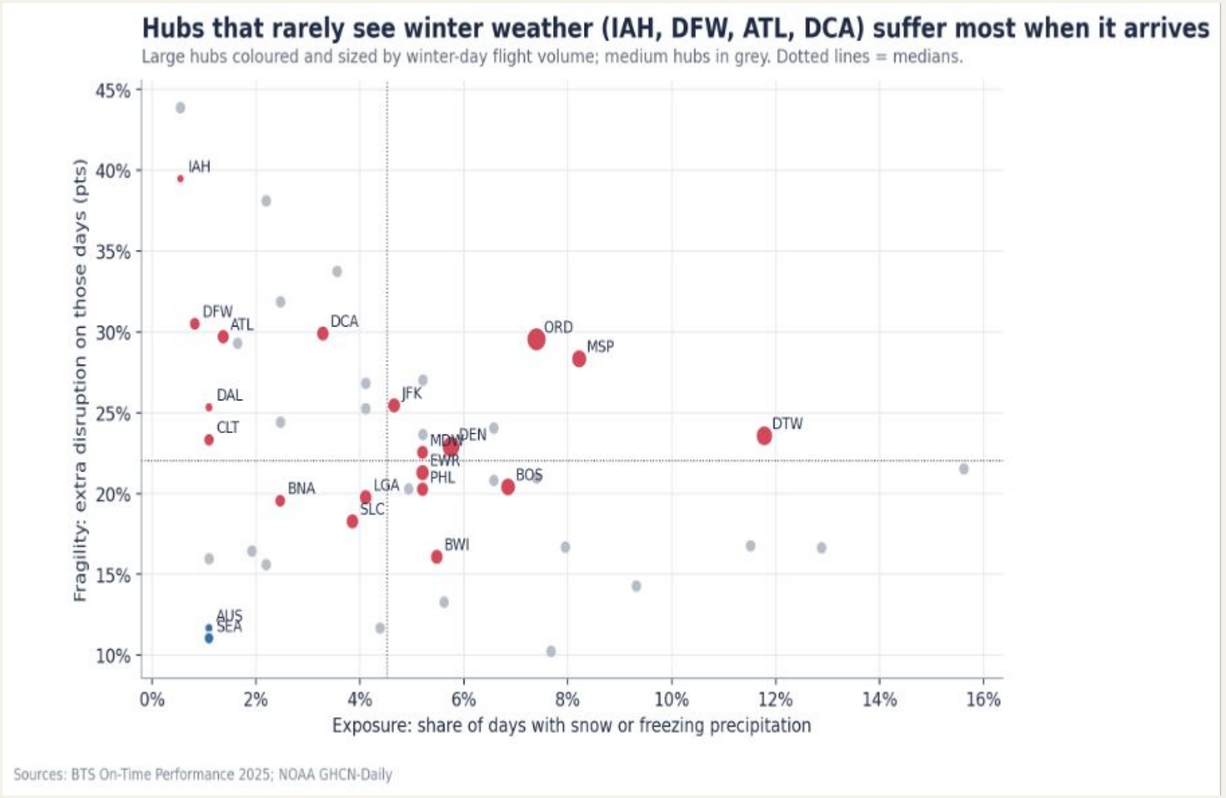


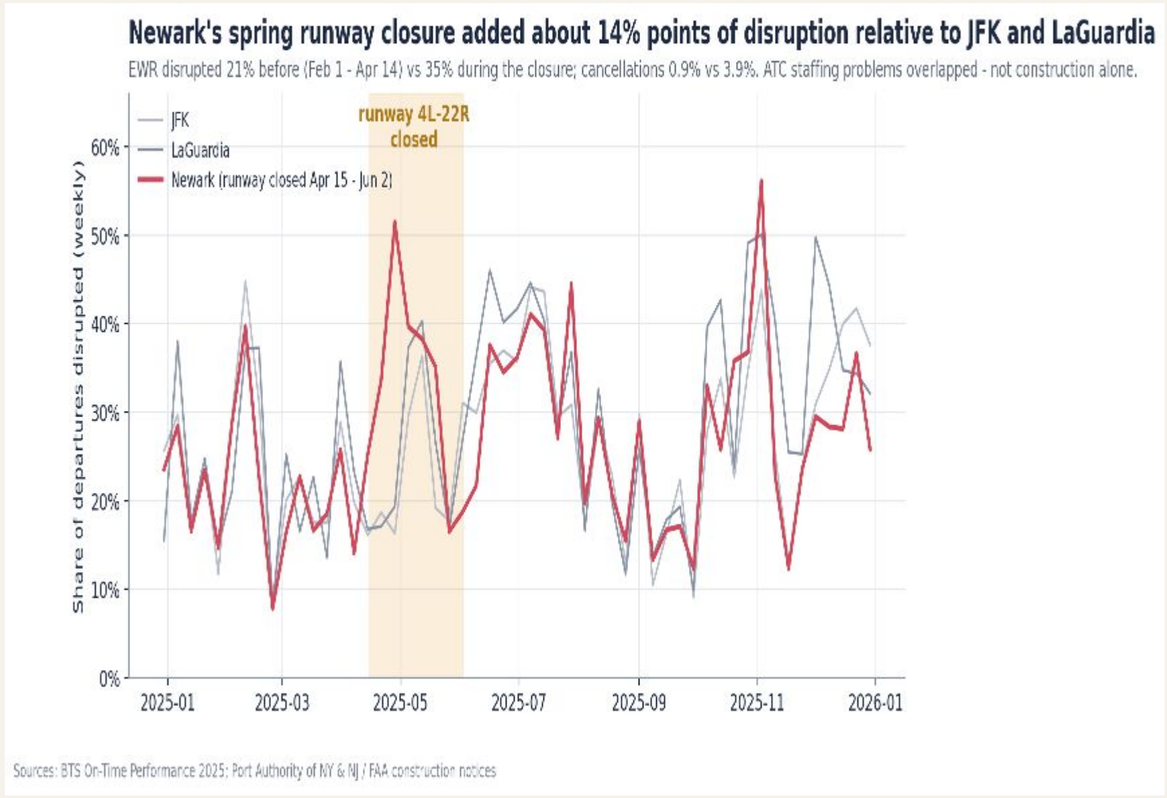
Appendix A4 — Planning tables: the buffer a must-attend traveler needs, and weather exposure by airport and month





Appendix A5 — Hub fragility: winter readiness by outcome, and Newark's 2025 runway closure





- Winter readiness is measured by outcome: how much a hub degrades on its own snow / freezing days. No public de-icing capacity list exists.
- Newark: 21% → 35% disrupted and 0.9% → 3.9% cancelled during the 4L-22R closure (15 Apr - 2 Jun 2025) while JFK and LGA were flat; ATC staffing problems overlapped. Hence a watch rule: check FAA construction notices before tight connections.
- Sources: BTS On-Time Performance 2025; NOAA NCEI GHCN-Daily; OurAirports; Port Authority of NY & NJ / FAA notices; full list in data/external/SOURCES.md.